Veritas Research Report: LLM-Driven Hardware Logic Generation

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1. Introduction

This report documents the experimental results of using the **Veritas** framework to generate and verify hardware logic via Large Language Models (LLMs). The primary goal was to explore the feasibility of generating Correct-by-Construction (CbC) hardware artifacts, such as CNF, BENCH, and Verilog, using automated prompting strategies.

2. Experimental Setup and Configurations

The experiment was conducted across three distinct design configurations to test the framework's versatility and scalability. In accordance with Part III of the assignment, **Gemini 2.5 Flash** was utilized as the primary reasoning engine, replacing the default GPT models.

Design Task	Type	Framework Goal	Status
Adder	3-bit	Model Transition (Gemini)	**Pass**
Multiplexer	2x1	Non-Adder Design (Part I)	**Pass**
Adder	5-bit	Scale-up Design (Part IV)	**Pass**

3. Part III: Model Transition and Observations

Switching to **Gemini 2.5 Flash** provided significant insights into API management and prompting logic.

- **Runtime Observation:** During the initial "Incremental Generation" phase, a 429 TooManyRequests error was encountered. This was identified as a Rate Limit restriction on the Gemini Free Tier, which limits RPM (Requests Per Minute).
- **Prompt Adjustment:** To mitigate this, I optimized the workflow from incremental prompting to **Single-shot Full-scale Generation**. This adjustment not only bypassed the API quota limits but also improved the logical coherence of the generated CNF by providing a strict variable mapping within a single context window.

4. Part IV: Scale-up Results

The design was scaled from a 3-bit Adder to a **5-bit Adder**.

- **Complexity Growth:** The number of clauses in the CNF file increased significantly, reflecting the added carry-propagation logic required for higher bit-widths.
- **Verification:** The 5-bit design successfully passed the Oracle verification, covering all possible input combinations. This demonstrates that with precise variable mapping, LLMs can handle medium-scale combinatorial logic.

5. Reflections on CNF as an Intermediate Representation

Through this assignment, I observed that **CNF (Conjunctive Normal Form)** serves as a robust intermediate representation for hardware logic because:

1. **Universality:** It represents pure Boolean constraints independent of specific hardware libraries.
2. **Verifiability:** It allows for direct formal verification and truth table simulation before committing to a gate-level Netlist (BENCH) or Verilog code.
3. **Hardware-Software Bridge:** It translates high-level functional descriptions into a format that SAT solvers and hardware synthesis tools can process efficiently.

6. Conclusion

The Veritas framework, coupled with the Gemini 2.5 Flash model, successfully automated the generation of verifiable hardware designs. Despite initial API limitations, the transition to single-shot prompting allowed for the successful creation and verification of a 5-bit Adder and a Multiplexer, fulfilling all requirements of the tutorial.